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C01 Assessment- 1

Concept Mapping



Course: CSA07 — Computer Networks | Questions 4, 5 & 6

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Course Name: Computer Networks

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Question 4

A concept map shows: Error Control → Data Link Layer → Error Detection → Transport Layer → Error Correction. Analyze how the interaction between these layers improves reliability in data transmission.

Introduction

Reliable data transmission over a network cannot be guaranteed by a single layer of the OSI model acting alone. Error control — the mechanism by which corrupted, lost, or duplicated data is identified and fixed — is distributed across multiple layers, each contributing a different piece of the reliability guarantee. Understanding how the Data Link Layer and the Transport Layer divide this responsibility is essential to understanding why layered network architectures are more robust than single-layer designs.

Problem Statement

Given that both the Data Link Layer and the Transport Layer perform error-control functions, the problem is to analyze how their individual roles (error detection at the link level and error correction at the end-to-end level) interact with one another, and to explain why this two-tier interaction produces a more reliable communication system than either layer could achieve on its own.

Concept Map

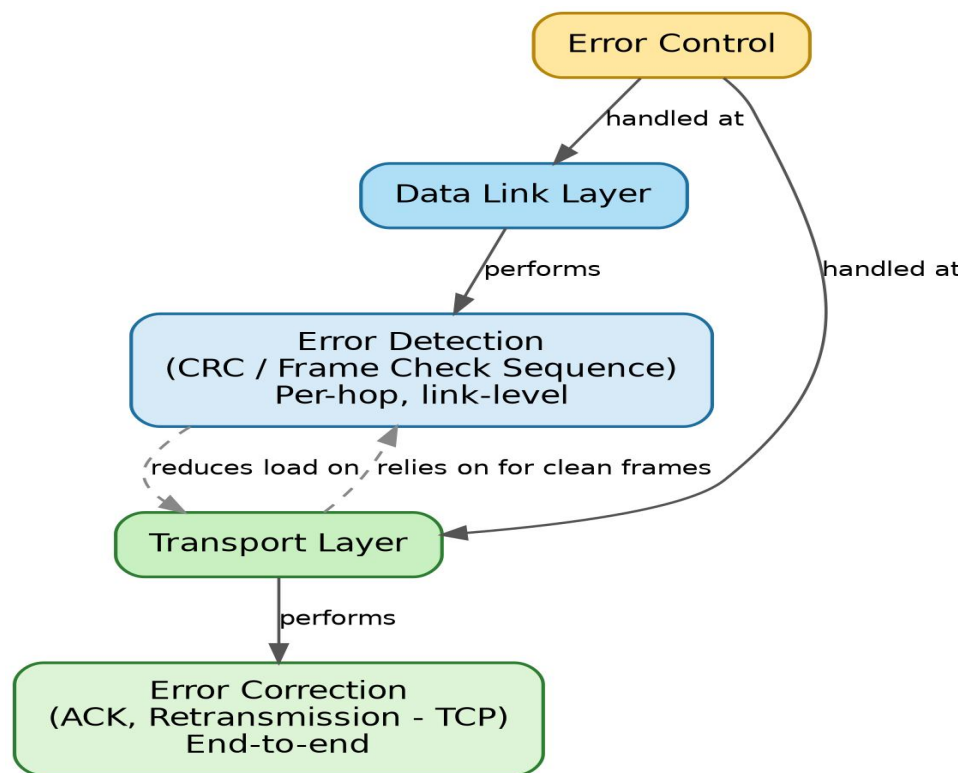


Figure 1: Concept map of error control across the Data Link and Transport layers

Theory and Analysis

The Data Link Layer is responsible for error detection on each individual physical link (hop) between two directly connected devices. It uses techniques such as the Cyclic Redundancy Check (CRC) or a Frame Check Sequence (FCS) appended to every frame. When a receiving node recalculates the checksum and finds a mismatch, it discards the corrupted frame immediately. This detection is local and fast, because it only concerns the single link the frame just traveled across — it has no visibility into the rest of the network path.

The Transport Layer, in contrast, is responsible for end-to-end reliability across the entire path from source to destination, which may span many links and intermediate routers. Protocols such as TCP use sequence numbers, acknowledgments (ACKs), timers, and retransmission to detect segments that never arrived, arrived out of order, or were corrupted beyond what the Data Link Layer could catch (for example, due to a queue overflow or routing failure at an intermediate hop).

- Data Link Layer error detection prevents obviously corrupted frames from being forwarded further, reducing wasted bandwidth and unnecessary load on upper layers.
- Transport Layer error correction compensates for failures the Data Link Layer cannot see — such as packet loss at a congested router — by using end-to-end acknowledgment and retransmission.
- The interaction is complementary, not redundant: link-level detection is fast and local, while transport-level correction is comprehensive and guarantees delivery across the whole path.

Because of this division of labor, the overall system achieves both efficiency and reliability: most errors are caught and discarded quickly at the point where they occur, while the small fraction of errors that escape link-level detection are still corrected end-to-end. This layered interaction is the reason data transmission over unreliable physical media can still deliver complete, accurate data to the application.

Question 5

Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

Introduction

Web browsing is one of the most common real-world applications of the OSI reference model. When a user opens a browser and requests a web page, data passes down through all seven layers on the client side, across the network, and back up through all seven layers on the server side. Each layer performs a distinct function, and the successful completion of every layer's task is a precondition for the layer above it to work.

Problem Statement

The problem is to map each stage of a typical web-browsing session onto its corresponding OSI layer, and then to analyze what happens to the overall communication process when any single layer in that chain fails — in order to demonstrate the interdependent nature of the OSI model.

Concept Map

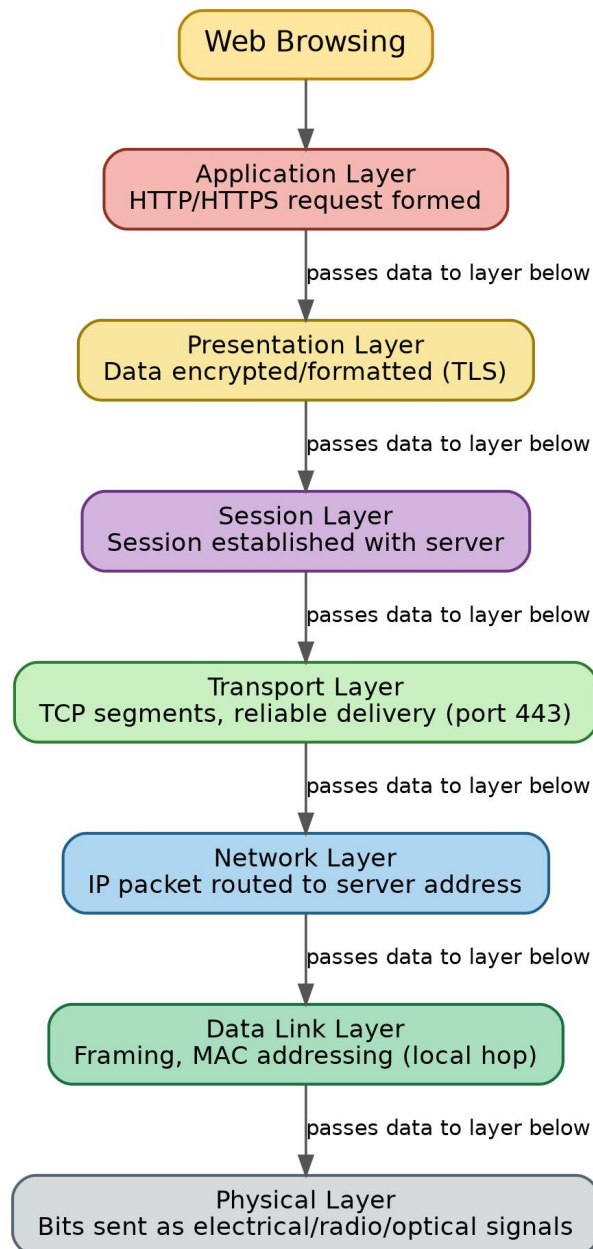


Figure 2: Concept map of a web browsing session mapped to the seven OSI layers

Theory and Analysis

A web browsing session moves through the OSI layers as follows: the Application Layer generates an HTTP/HTTPS request; the Presentation Layer formats and encrypts the data (e.g., via TLS); the Session Layer establishes and manages the session with the web server; the

Transport Layer breaks the data into TCP segments and ensures reliable delivery over a specific port (443 for HTTPS); the Network Layer wraps segments into IP packets and routes them toward the server's IP address; the Data Link Layer frames the packets and handles local addressing via MAC addresses; and the Physical Layer converts the frames into electrical, optical, or radio signals that travel across the transmission medium.

Because each layer relies on the successful output of the layer beneath it, the OSI model behaves like a dependent chain rather than a set of independent components. A failure at any point breaks the entire process:

- Physical Layer failure (e.g., unplugged cable, no Wi-Fi signal): no bits are transmitted at all, so nothing above this layer can function — the page will not even begin to load.
- Data Link/Network Layer failure (e.g., misconfigured IP address, routing/DNS failure): frames or packets cannot reach the destination, producing errors such as 'server not found' or 'destination unreachable.'
- Transport Layer failure (e.g., blocked port, firewall rule): the connection cannot be established or maintained, resulting in a connection timeout even though lower layers are functioning.
- Application Layer failure (e.g., the web server itself is down): a connection may form successfully, yet no valid HTTP response is returned, producing an error page.

This analysis shows that reliable web browsing depends on the correct, sequential operation of all seven layers. A malfunction anywhere in the stack propagates upward and disrupts the entire communication process, which is precisely why the OSI model is described as a layered and interdependent architecture.

Question 6

A network issue is represented in the concept map: Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗. Analyze the problem and explain how the concept map helps in troubleshooting.

Introduction

Network troubleshooting is most effective when it follows a structured, layer-by-layer methodology rather than a random search for the fault. The OSI model provides exactly this structure: because each layer depends on the one below it, testing layers from the bottom up allows a technician to isolate precisely where a failure originates, rather than guessing across the whole system.

Problem Statement

Given a scenario where the Physical and Data Link layers are confirmed working but the Network, Transport, and Application layers all show failure, the problem is to identify the most probable root cause of the fault and to explain how representing this status as a concept map supports efficient, targeted troubleshooting.

Concept Map

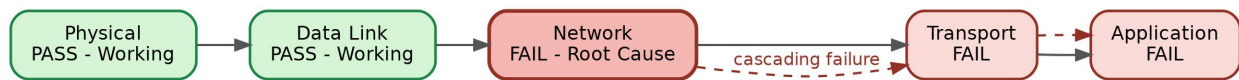


Figure 3: Concept map of layer-wise status showing the fault originating at the Network layer

Theory and Analysis

The concept map shows the Physical Layer and Data Link Layer functioning correctly — this means the cable/wireless signal is present and local frame delivery (e.g., link connectivity, ARP resolution, switching) is working normally within the local segment. The failure begins at the Network Layer, which is responsible for logical addressing (IP addresses) and routing packets toward their destination. A fault here typically points to causes such as an incorrect IP configuration, an unreachable default gateway, or a broken routing path.

Because the Transport and Application layers depend entirely on the Network Layer to deliver packets to the correct destination, their failure is a direct consequence of the Network Layer fault rather than a separate, independent problem. If the Network Layer cannot route packets, the Transport Layer has nothing to acknowledge or deliver, and the Application Layer never receives a response.

- Root cause identification: the concept map immediately highlights the Network Layer as the point where the chain of success breaks, which is where diagnostic effort (e.g., checking IP configuration, gateway reachability, or routing tables) should be focused.
- Avoiding wasted effort: since Physical and Data Link are confirmed working, there is no need to re-check cabling, switch ports, or link lights — this saves diagnostic time.
- Understanding cascading failure: the map visually demonstrates why Transport and Application also show failure — they are downstream symptoms, not independent faults.

This demonstrates the core practical value of a concept map in networking: it converts a vague symptom ('the internet is not working') into a precise, layer-located diagnosis, enabling a systematic, bottom-up troubleshooting approach — typically using tools such as ping (Network Layer reachability) or traceroute — rather than an inefficient trial-and-error search.